

Design Document — Personal Homepage

Author: Ayush Nimeshkumar Patel **Course:** [CS5610 Web Development](#), Northeastern University
Project: Project 1 — Your personal home page **Live site:** <https://ayushp2207.github.io/ayush-homepage/> **Repository:** <https://github.com/ayushp2207/ayush-homepage> **PDF version:** [DESIGN.pdf](#)

1. Project description

What this is

A static personal homepage built with vanilla HTML5, CSS3 and ES6 modules. No backend, no framework, no component library, no build step. Bootstrap 5 is used for its grid only; every other style is hand-written.

The problem it solves

A résumé is a list of claims. "Kubernetes, PyTorch, LangChain" tells a reader that I have touched those things, but not *where*, *when*, or *how seriously*. Recruiters skim for keywords; engineers want evidence. The two audiences need the same facts presented at different depths, and a flat list of bullet points serves neither well.

So the organising idea of this site is **evidence, not adjectives**. Every skill claim is wired to the specific role, project or paper where I used it, and the site makes that wiring browsable rather than asking the reader to reconstruct it from prose.

Scope

Three pages:

Page	URL	Purpose
Home	<code>index.html</code>	Identity, the skill constellation, experience, education, toolkit
Projects	<code>projects.html</code>	Longer write-ups of projects and publications

Page	URL	Purpose
Built with AI	<code>built-with-ai.html</code>	The assignment's AI-generated page, used as a full GenAI disclosure

Non-goals

- No blog or CMS. Content lives in one JavaScript module and is edited by hand.
 - No contact form, because that would need a backend.
 - No dark/light toggle. The site commits to one dark palette and does it well.
-

2. User personas

Persona 1 — Priya, technical recruiter

- **Role:** University recruiter at a cloud infrastructure company.
- **Context:** Screening ~120 candidates for summer internships. Roughly 40 seconds per profile on the first pass, on a laptop with a dozen tabs open.
- **Goals:** Confirm graduation date, work authorisation timing, and whether the candidate has shipped anything at production scale.
- **Frustrations:** Portfolios that open with a full-screen animation and bury the graduation date three scrolls down. Having to open a PDF to learn anything concrete.
- **What she needs from this site:** Name, current role, graduation date and availability visible without scrolling. One click to email.

Persona 2 — Daniel, engineering hiring manager

- **Role:** Senior engineer who will run the technical interview.
- **Context:** Already read the résumé and is deciding whether to spend an hour on a phone screen. Reads carefully, and is sceptical of inflated claims.
- **Goals:** Find out whether the candidate's "multi-agent systems" experience is a tutorial project or real work; see actual code.
- **Frustrations:** Skill bars showing "Python 90%". Project cards with a title, a logo and no explanation of what was actually built or how hard it was.
- **What he needs from this site:** The ability to pick a technology and immediately see every place it was used, with enough detail to form a question. A link to source.

Persona 3 — Meera, fellow student and peer reviewer

- **Role:** Classmate assigned to review this submission.
 - **Context:** On a phone, between classes, working through a rubric.
 - **Goals:** Check the rubric items — semantic markup, responsive layout, alt text, original JS, MIT license, README quality.
 - **Frustrations:** Sites that only work at desktop width. Creative additions that turn out to be a copied library demo.
 - **What she needs from this site:** A layout that holds up at 390px, a clearly identified creative addition, and an honest statement of what was generated.
-

3. User stories

Written as stories with acceptance criteria, so each one is testable.

Epic A — Establish identity fast

A1. *As Priya, screening quickly, I want the candidate's name, current role and availability without scrolling, so I can triage in seconds.*

- Given a 1440×900 desktop viewport
- When the homepage loads
- Then the name, a one-sentence summary, and "Available ... starting Summer 2027" are all visible above the fold
- And an email link is reachable in one click

A2. *As Priya, I want to know where the candidate studies and when they graduate, so I can match them to a requisition.*

- Given the homepage
- When I reach the Education section
- Then each degree shows institution, degree name, expected date and location

Epic B — Show how I actually think

B1. *As Daniel, I want to ask about a technology in my own words and find the relevant work, rather than scanning a list.*

- Given the query router on the homepage
- When I type "multi-agent orchestration over MCP"
- Then it routes to the AT&T system
- And it lists the terms that carried the score with their contributions

B2. *As Daniel, I want to know whether the match was close or decisive.*

- Given any confident result
- Then the runner-up documents are listed with their scores

B3. *As Daniel, I want it to admit when it doesn't know.*

- Given a query unrelated to my work
- Then it reports "no confident match" rather than the nearest document
- And names the query terms absent from the corpus

B4. *As Meera on a phone, I want to try it without typing.*

- Given the example chips below the input
- When I activate one
- Then that query runs immediately

B5. *As Daniel, I want to find where a given skill was used.*

- Given the skill constellation on the Projects page
- When I select "Kubernetes"
- Then every role and project using it is listed, and unrelated nodes dim

Epic C — Depth on demand

C1. *As Daniel, I want longer write-ups than a résumé bullet, so I can prepare questions.*

- Given the Projects page
- Then each entry has a summary plus specific detail points
- And publications are listed with venue and status

Epic D — Work everywhere, for everyone

D1. *As Meera on a 390px phone, I want a usable layout.*

- Given a 390px viewport

- Then the page never scrolls horizontally
- And the navigation collapses behind a labelled Menu button
- And the constellation still renders and responds

D2. *As a keyboard user, I want to skip repeated navigation.*

- Given focus at the top of the document
- When I press Tab once
- Then a "Skip to main content" link appears and is focusable

D3. *As a screen reader user, I want the graph not to be a dead end.*

- Given the canvas
- Then it exposes `role="img"` and a description saying the same nodes exist as buttons below

D4. *As someone with vestibular sensitivity, I don't want a graph animating at me.*

- Given `prefers-reduced-motion: reduce`
- Then the layout is computed without animating and painted once

Epic E — Academic honesty

E1. *As the grader, I want to know exactly which parts were AI-generated.*

- Given the "Built with AI" page
- Then it names the model and version, lists what was asked, separates generated from hand-authored content, and records where the model was wrong

4. Information architecture

```
index.html ..... Home
├── Hero ..... name, one-line pitch, contact, availability
├── Query router ..... the creative addition
├── Experience ..... reverse-chronological timeline (4 roles)
├── Education ..... 2 degrees
├── Toolkit ..... grouped skill lists
projects.html ..... Projects + Publications + skill constellation
built-with-ai.html ..... GenAI disclosure (the AI-generated page)
```

All three pages share one persistent header and footer. Content is defined once in `js/data.js`; the constellation and the rendered sections both read from it, so a role is never described in two

places.

5. Design mockups

Wireframes produced before implementation. Final screenshots follow in section 8.

5.1 Home — desktop (≥ 992px)

[AP] Ayush PatelHomeProjectsBuilt with AIGitHub|sticky

BOSTON, MA
Ayush Nimeshkumar Patel<- h1, clamp()
MS CS at Northeastern, currently building control-plane
services at AWS. Multi-agent AI systems and the
infrastructure that makes them reliable.

(See my work) (email) (LinkedIn) (GitHub)<- pill links
| Available for full-time roles starting Summer 2027<- accent rule

SKILL CONSTELLATION
A résumé tells you someone knows Kubernetes. It rarely tells
you where they used it.

Databricks

DNS Agents

Python

Pit Wall

Rand-PC

MCP

Pibit

5G Testbed

● Skill● Experience● Project● Publication

PICK A NODE
(AWS)(Computer Vision)(Databricks)
(Deep RL)(DynamoDB)(Java)(Python)
(Kubernetes)(LangChain)(MCP) ...
(Clear selection)

7 cols

Python
7 PLACES I HAVE USED IT
– DNS Anomaly Detection
– 5G Spectrum Testbed
– Pit Wall

5 cols

EXPERIENCE
● SEP 2026 – PRESENT
| Software Development Engineer Intern
| Amazon Web Services · Boston, MA
| Designing S3 endpoint configuration for Transfer Family...
● JUN 2026 – AUG 2026
| AI Engineer Intern ...

EDUCATION

TOOLKIT

© 2026 Ayush Patel · Built for CS5610 · Home Projects Source

5.2 Home — mobile (< 768px)

Single column throughout. Nav collapses; the constellation keeps its full interaction but shortens to 340px tall; the detail panel stacks under the chips.

[AP] Ayush Patel (Menu)

BOSTON, MA
 Ayush Nimeshkumar Patel
 MS CS at Northeastern,
 currently building...
 (See my work)
 (email) (LinkedIn)
 | Available Summer 2027

SKILL CONSTELLATION

●Skill ●Exp ●Proj

PICK A NODE

(AWS)(CV)(Databricks)

(Deep RL)(DynamoDB) ...

Python
 7 PLACES

EXPERIENCE

● SEP 2026 – PRESENT
 | SDE Intern
 | Amazon Web Services

Nav open state:

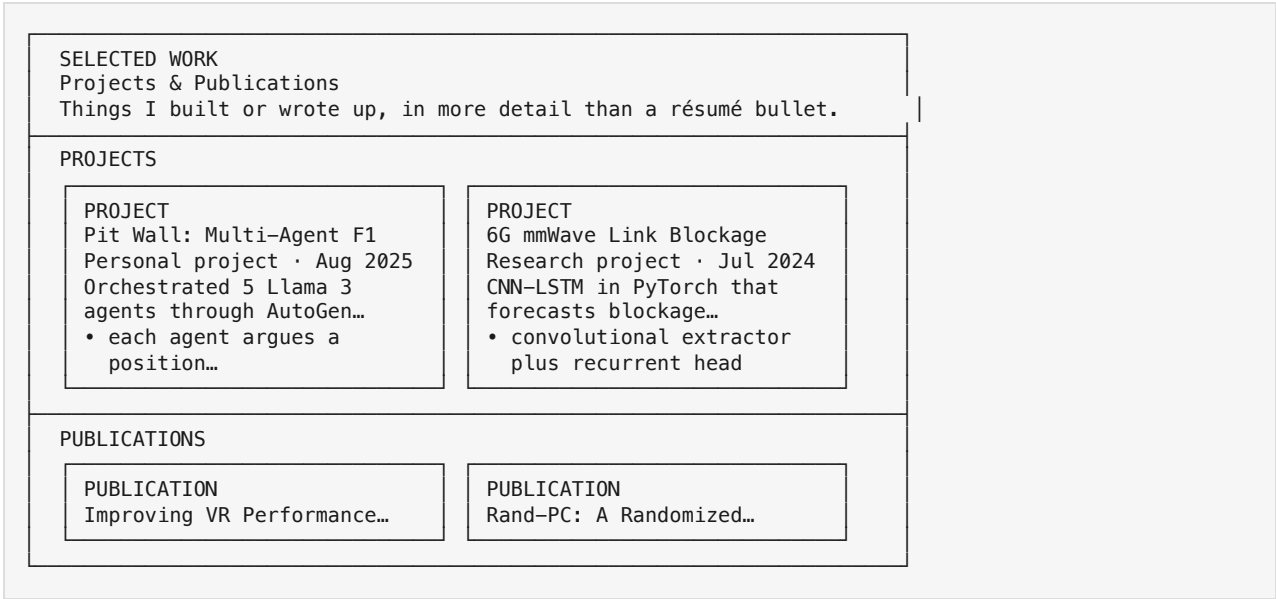
[AP] Ayush Patel (Menu)

Home
 Projects
 Built with AI
 GitHub

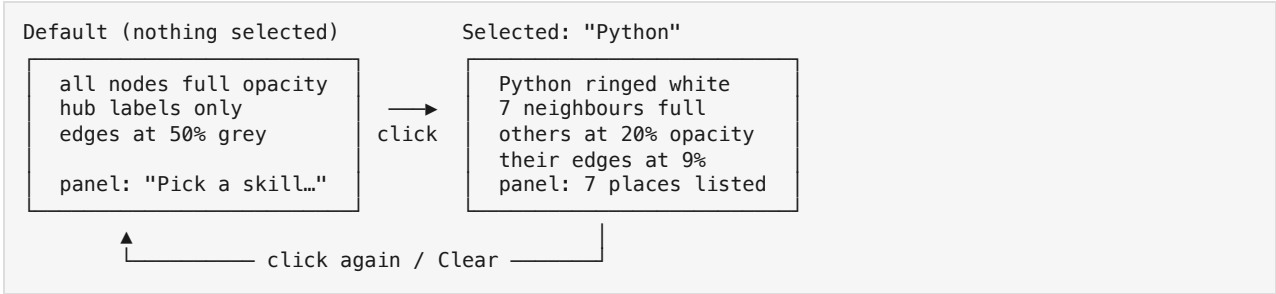
aria-expanded="true"

5.3 Projects — desktop

Two-column card grid (col-12 col-lg-6), collapsing to one column on mobile.



5.4 Constellation interaction states



6. Design system

Palette

Warm near-black rather than blue-black, with a single bright green carrying every interactive element. Only one hue does the accent work: everything interactive is green and nothing decorative is, which is what keeps a dark page from turning muddy.

Token	Value	Use
--ink	#262421	Page background (warm black)
--surface	#312e2b	Cards, nav, status panel
--surface-raised	#3d3a37	Pills, chips, buttons

Token	Value	Use
--line	#4a4642	All borders and rules
--text	#f1efed	Body text
--text-muted	#a9a29b	Secondary text
--accent	#81b64c	Links, primary buttons
--board-light	#eeeeed2	Light squares
--board-dark	#769656	Dark squares
--board-*--active	#f6f669 / #baca2b	Last-move highlight
--warn	#e0a63c	"No preference" notices

Only one hue does the accent work. Everything that is interactive is green; nothing decorative is. That is what keeps a dark page from turning muddy.

Type and spacing

System font stack, so there is no web-font round trip. The h1 uses `clamp(2.1rem, 6vw, 3.4rem)` to scale without breakpoints. Body copy is capped at 58–70ch for readability. Spacing is a six-step scale (0.25/0.5/1/1.5/2.5/4rem) exposed as custom properties, so vertical rhythm stays consistent.

CSS conventions

- Every element is targeted by **class**, never by id, and never by bare tag beyond base resets.
- **Zero !important** in the stylesheet (verified: `grep -c '!important' → 0`).
- Layout uses Bootstrap's grid plus CSS flexbox and grid. One deliberate override: `.page-wrap` `.row` zeroes Bootstrap's negative gutters, because `.page-wrap` is not a Bootstrap `.container` and the negative margins otherwise cause horizontal overflow on mobile.

Accessibility decisions

Concern	Decision
Keyboard access to the graph	Every node is also a real <code><button></code> with <code>aria-pressed</code>
Screen readers	Canvas is <code>role="img"</code> with a label pointing at the button list
Skip navigation	<code>.skip-link</code> , off-screen until focused

Concern	Decision
Focus visibility	One shared <code>:focus-visible</code> outline, 3px, offset 3px
Motion sensitivity	<code>prefers-reduced-motion</code> settles the layout without animating
Semantic controls	Buttons are <code><button></code> , links are <code><a></code> — no clickable <code><div></code>
Images	Every <code></code> carries a real <code>alt</code>

7. The creative addition: a retriever over my own work

Why this and not a honeycomb grid

The assignment's own example is a honeycomb image grid, which is therefore the one thing guaranteed to appear in other submissions. More importantly, a decorative grid says nothing about how I work.

Most of what I actually do is routing. At AT&T a parent orchestrator picks which of three specialist sub-agents handles a request over MCP. At the WiNES lab I built MCP servers exposing hardware as LLM-callable tools, then measured how often the model picked the right one. At Pibit I built the retrieval layer deciding which documents an answer is built from.

All three are the same problem — score candidates against a query, pick one, defend the choice — so the widget is that, pointed at my own résumé.

Mechanism

`js/retriever.js`, with no search library:

- Tokenise, drop stopwords, suffix-stem
- **BM25** ranking, $k_1 = 1.5$, $b = 0.75$
- Smoothed IDF: $\ln(1 + (N - df + 0.5) / (df + 0.5))$, always positive, so a matching term can never count *against* a document — which would be impossible to explain to a reader looking at the breakdown
- Per-term contribution tracking, which is what makes the explanation possible

What the interface exposes

Shown	Why
Terms that carried the score, with values	Makes the ranking arguable rather than magic
Runner-up documents with scores	Shows how close the call was
Query terms absent from the corpus	Makes a miss explainable
An explicit "no confident match"	See below

The refusal is the point

A retriever that always returns its best guess is the most common way tool routing fails in production: it answers confidently out of the wrong document. Below a BM25 floor of 0.9 this one says so instead.

Two bugs the design exposed

Short-document bias. The first version scored cosine similarity over TF-IDF vectors, and "multi-agent orchestration over MCP" routed to *Pit Wall* rather than the *AT&T* system. Both are genuinely multi-agent, but Pit Wall's description is shorter and L2 normalisation over-rewards short documents. BM25 normalises by length relative to the corpus average, which is precisely the problem it was designed for.

A missing stemmer rule. Even after BM25 the same query still missed. The résumé says "orchestrator"; the query says "orchestration". The stemmer collapsed -ation, -ating and -ated to a shared stem but had no rule for the -ator agent noun, so the two forms never met. One rule fixed it, and the query then routed correctly.

Routing accuracy

12 of 12 on a held-out set of realistic questions, including two that must return nothing. Reproducible with `npm run test:routing`.

Query	Expected
multi-agent orchestration over MCP	AT&T DNS agents
how do you keep data consistent under concurrent writes?	AWS Transfer Family
computer vision on video	6G mmWave

Query	Expected
retrieval augmented generation embeddings	Pibit.ai
kubernetes and openshift	WiNES 5G testbed
reinforcement learning	VR WiFi-6 paper
formula 1 race strategy	Pit Wall
bayesian networks	Rand-PC
anomaly detection in DNS logs	AT&T DNS agents
what did you build at amazon?	AWS Transfer Family
tell me about your favourite pizza topping	no confident match
blockchain smart contracts	no confident match

7b. Second interactive piece: Skill Constellation

On the Projects page. A force-directed graph wiring all 23 skills and work items together: pick a skill and every place I used it stays lit while the rest dims.

Fruchterman–Reingold: repulsion k^2/d , attraction d^2/k , ideal distance $k = 0.62 \cdot \sqrt{(\text{area} / \text{nodeCount})}$ derived from canvas size and node count, and a cooling temperature so it converges. Labels are drawn in a second pass with rectangle-collision detection.

8. Final implementation

Home — hero

**Ayush Patel**

Home Projects Built with AI GitHub

BOSTON, MA

Ayush Nimeshkumar Patel

MS Computer Science student at Northeastern, currently building control-plane services at AWS. I spend most of my time on multi-agent AI systems, the infrastructure that makes them reliable, and the networks they run over.

[See my work](#)ayushnpatelworks@gmail.com[LinkedIn](#)[GitHub](#)

Available for full-time software engineering roles starting **Summer 2027**.



At the AT&T Innovation Center R&D Labs, Middletown NJ, during my Network CTO internship.

ASK MY WORK A QUESTION

Most of what I do is routing: picking which specialist agent or tool should handle a request, and being able to defend the choice afterwards. This is that, pointed at my own résumé.

Type a question and a **BM25 retriever** scores it against every role, project and paper below, then shows you the terms that carried the score. If nothing scores well enough it says so, rather than confidently handing you the nearest document — which is the standard behavior of a naive retriever.

The query router

Ayush Patel

Home

Projects

Built with AI

GitHub

multi-agent orchestration over MCP

how do you keep data consistent under concurrent writes?

computer vision on video

retrieval augmented generation

reinforcement learning

Routed to

Multi-Agent DNS Anomaly Detection

EXPERIENCE

AT&T Labs, Network CTO · Jun 2026 – Aug 2026

Architected a multi-agent AI system for DNS anomaly detection and root cause analysis: a parent orchestrator coordinating 3 specialized sub-agents over MCP, processing 8M+ residential gateway logs to diagnose network faults end to end.

TERMS THAT CARRIED THE SCORE

agent 1.86

mcp 1.70

multi 1.55

orchestrat 1.13

ALSO CONSIDERED

Pit Wall: Multi-Agent F1 Strategy Simulation

Agentic 5G Spectrum Testbed

BM25 over 9 documents, 4 query terms, best score 6.24. [See all projects](#)

EXPERIENCE

SEP 2026 – PRESENT

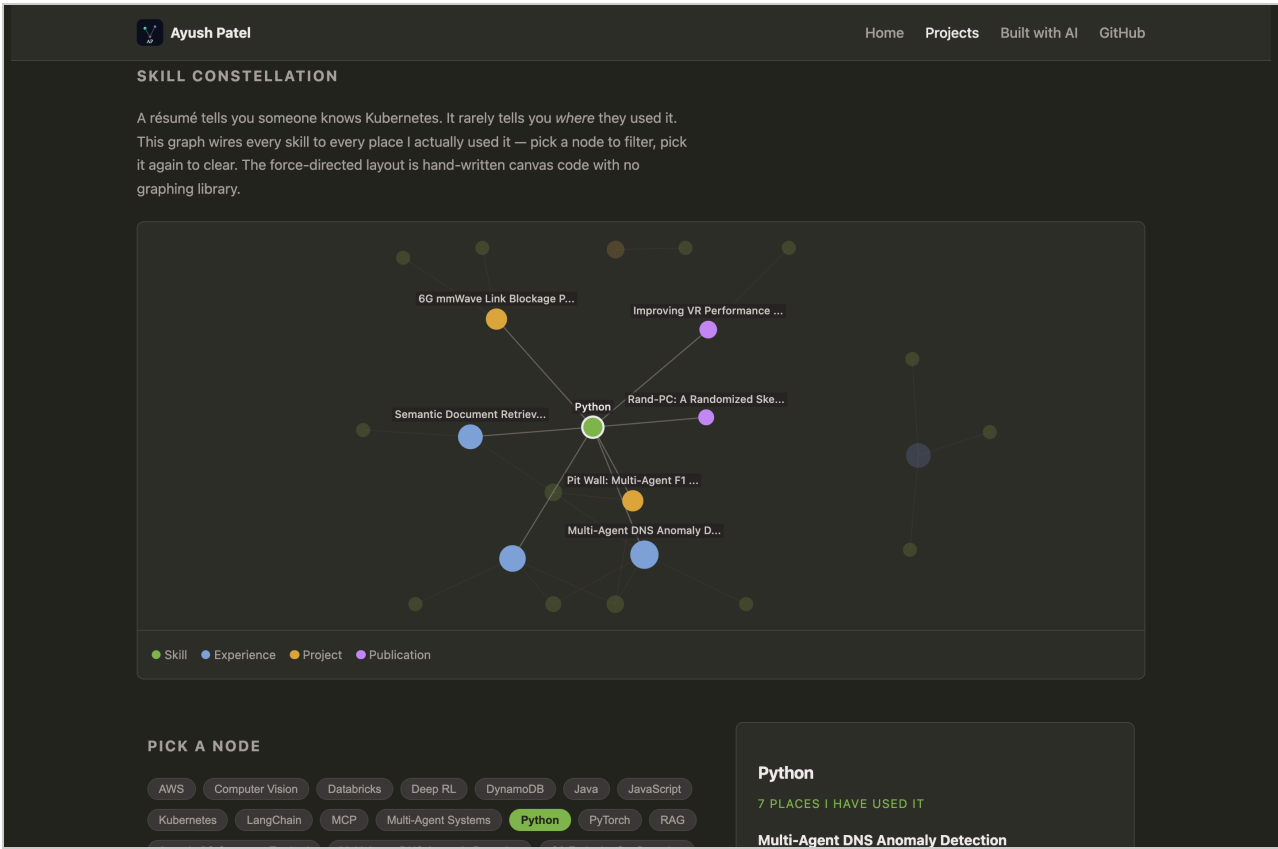
Software Development Engineer Intern

Amazon Web Services · Boston, MA

Designing S3 endpoint configuration for AWS Transfer Family Web Apps, so customers can expose their own S3 buckets through a managed browser interface without provisioning SFTP, FTPS, FTP or AS2 clients.

The result names the routed project, the terms that carried the score with their BM25 contributions, and the runners-up it beat.

Skill constellation, Projects page



Mobile



Ayush Patel

Menu

BOSTON, MA

Ayush Nimeshkumar Patel

MS Computer Science student at Northeastern, currently building control-plane services at AWS. I spend most of my time on multi-agent AI systems, the infrastructure that makes them reliable, and the networks they run over.

[See my work](#)

ayushnpatelworks@gmail.com

LinkedIn

GitHub

Available for full-time software engineering roles starting **Summer 2027**.



9. Verification

Check	Method	Result
W3C validity	validator.w3.org/nu on all 3 pages	0 errors
ESLint (class config)	npx eslint .	0 errors
Prettier	npx prettier --check	clean
Console errors	Headless Chrome, all 3 pages	none
Images have alt	DOM audit	0 missing
Horizontal scroll at 390px	scrollWidth vs innerWidth	none
Keyboard filtering	Tab + Enter on node list	works
!important count	grep -c on the stylesheet	0

